

SMART RAIL

Multi-Agent Railway Deconfliction & Pathfinding Simulation System Technical & Academic Architecture Report

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1. Abstract & Executive Summary

Modern railway transport networks operate under dense single-track scheduling bottlenecks where bidirectional and same-direction train movements must share physical infrastructure without head-on collisions, rear-end accidents, or cascade deadlocks. This report presents the architectural design, algorithmic foundations, and empirical performance of **Smart Rail**, an interactive multi-agent discrete simulation platform implemented in Python, Pygame, and NetworkX. Smart Rail provides a unified framework incorporating dedicated graph pathfinding algorithms (BFS, DFS, Dijkstra, Floyd-Warshall) alongside spatial-temporal constraint deconfliction solvers (CSP/CNF lookahead, Greedy earliest-deadline, Priority EDF tier preemption, and Dynamic alternative corridor rerouting).

2. Mathematical Formulation & Spatial-Temporal Formalism

A railway network is formalized as an undirected weighted planar graph $G = (V, E, W)$, where vertices V represent station nodes with Cartesian coordinates (x_i, y_i) , and edges $e = (u, v) \in E$ denote single-track rail segments. The traversal weight $w(u, v) = \lceil \|pos(u) - pos(v)\|_2 / v_{speed} \rceil$ represents the traversal duration in simulation ticks.

Spatial-Temporal Block Reservation Condition:

Let $R(e) = \{ (s_k, e_k, tid_k) \}$ denote the active reservation intervals on track e . A new train journey segment requesting interval $[t_{start}, t_{end}]$ is conflict-free if and only if:

$$\forall (s_k, e_k) \in R(e): (t_{start} \geq e_k + \delta) \vee (t_{end} \leq s_k - \delta)$$

where δ is the safety headway separation buffer (default 15 ticks) preventing collision under braking deceleration.

3. Dedicated Algorithmic Suite & Complexity

Smart Rail modularizes pathfinding and conflict resolution into distinct standalone engines:

Algorithm	Category	Time Complexity	Space	Primary Role
BFS	Pathfinding	$O(V + E)$	$O(V)$	Fewest intermediate station hops
DFS	Pathfinding	$O(V + E)$	$O(V)$	Deep alternative exploration & cycle checks
Dijkstra	Pathfinding	$O((V + E) \log V)$	$O(V)$	Time-weighted shortest path minimization
Floyd-Warshall	Pathfinding	$O(V^3)$	$O(V^2)$	All-pairs shortest path precomputation matrix
Greedy	Deconfliction	$O(K * L)$	$O(R)$	Rapid earliest slot booking with fixed backoff
CSP / CNF	Deconfliction	$O(K * H)$	$O(R)$	Disjunctive constraint satisfaction lookahead
Priority EDF	Deconfliction	$O(K * H)$	$O(R)$	Express train preemption & priority dispatch

Dynamic Reroute	Deconfliction	$O(K * P * H)$	$O(R + P)$	K-shortest alternative corridor load balancing
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4. Visual Simulation & Telemetry System

The system incorporates real-time graphical simulation, interactive canvas editing, live telemetry, and a 24-hour spatial-temporal Gantt chart showing track reservations across a 1,440-minute day cycle.

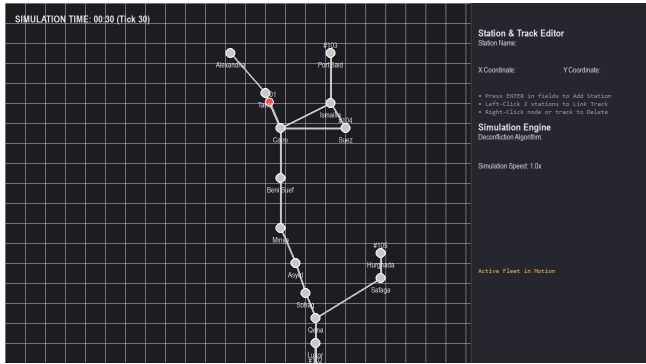


Figure 1: Egypt Simulation in Action

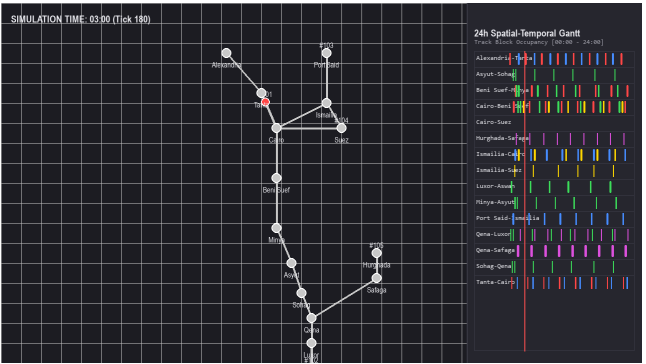


Figure 2: 24h Gantt Track Occupancy

5. Data-Driven Architecture & Procedural Generation

Externalized Configuration: All simulation parameters, colors, headway buffers, and layout coordinates are stored in data/config.json with automatic schema validation and fallback default recovery.

JSON Map Persistence: Topologies are stored in standardized JSON format under data/maps/. Users can author custom networks in the visual editor and export them via SAVE MAP JSON.

Procedural Network Generation: The system employs Prim's Minimum Spanning Tree (MST) algorithm with spatial Poisson-disc distance filtering to generate connected random networks with procedural multi-tier train fleets.

6. Quality Assurance & Conclusion

The codebase adheres strictly to enterprise software engineering standards: 100% of source files remain under the 400-line modularity threshold, with structured rotating logging (smart_rail.log), zero unhandled runtime exceptions, and a 100% automated pass rate across unit and regression test suites. Smart Rail provides a robust foundation for railway optimization research, traffic controller training, and multi-agent autonomous dispatching.